Biped Walking Robot

BY GONE RAMU

Project Aim

- To make a 6 DOF Biped Walking Robot using Arduino Mega.

Apparatus Required

Hardware Tools :

1. Arduino Mega
2. Servo Shield
3. 6 Servo Motors
4. Jumper wires
5. Arduino & USB cable
6. 2 Battery Source
7. Screws and nuts
8. Clamps

Apparatus Required

Software Tools :

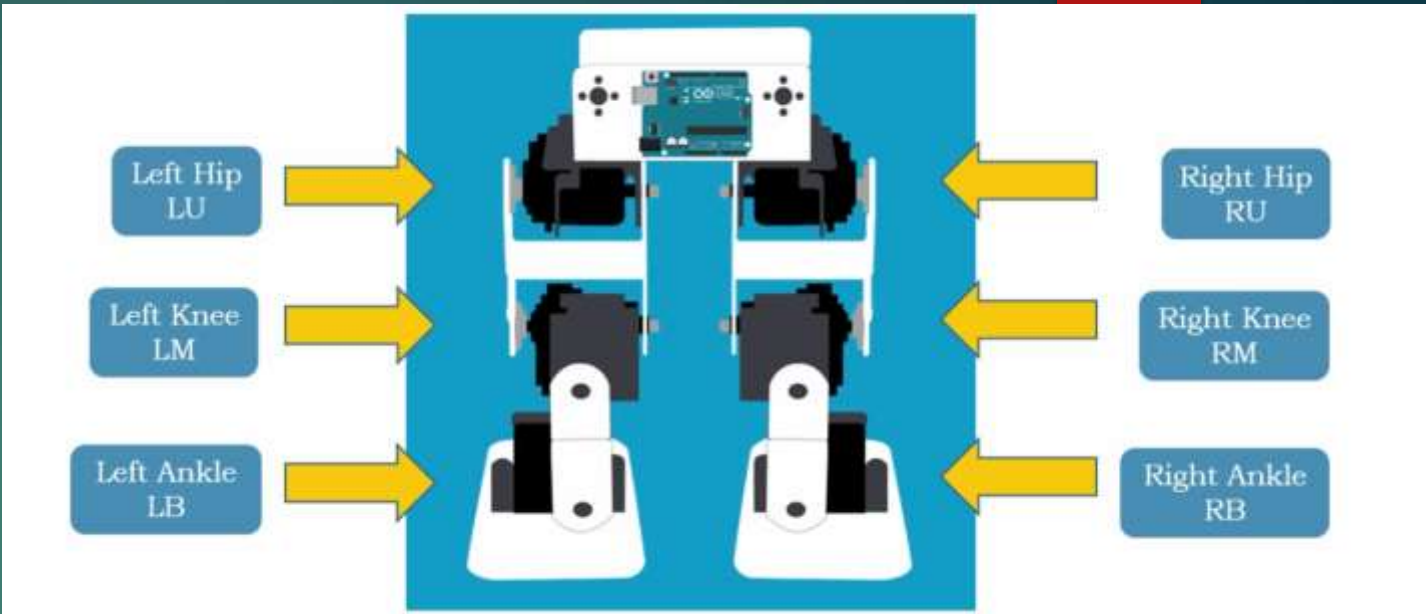
1. Arduino IDE
2. Adafruit library
3. Arduino code of Biped Walking Robot
4. Biped Robot cad file

Procedure

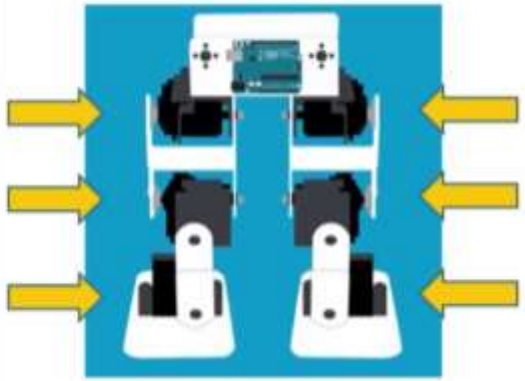
- Arrange 6 Servos in according to Biped Walking Robot structure along with its parts.
- Integrate 6 Servos with Servo Shield.
- Now connect the 6 Servos and Servo shield to Arduino mega.
- Connect Batteries to Servo shield for supplying the power.
- Plan the Walking Strategy for Biped Robot.
- After Gait planning , Open the Arduino IDE
- Upload the Biped Walking Robot Code in to Arduino mega.
- Biped Walking Robot is Done.

Circuit Connection

- 6 Servos positions in Biped Robot and Their Labels :



Component 1		Channel	Component 2
Servo LB	→	0	Servo Driver Shield
Servo LM		2	
Servo LU		4	
Servo RU		6	
Servo RM		8	
Servo RB		10	



- 6 Servos connection with Servoshield

Circuit Connection





Servoshield connection with Batteries :

Servoshield	Battery
VCC	Anode
GND	Cathode/GND

Circuit Connection

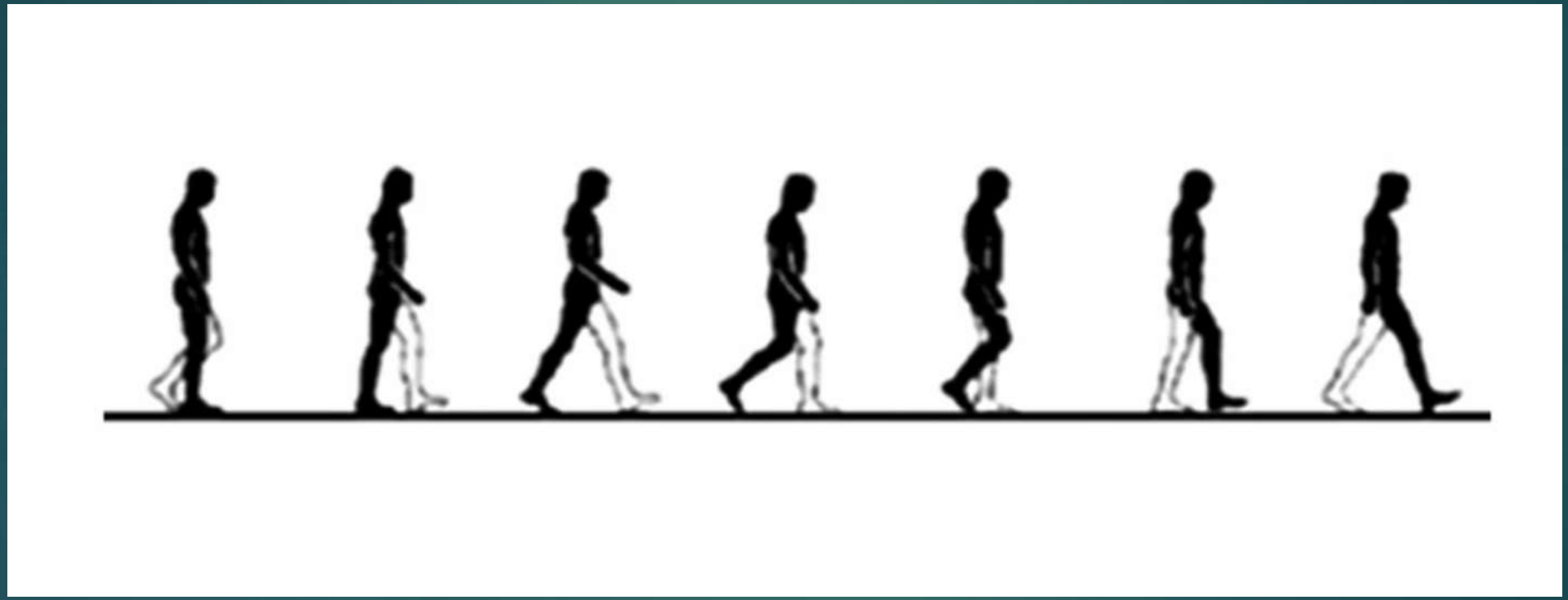




Servoshield connection with Arduino Mega :

Servoshield	Arduino mega
VCC	5v
GND	GND
SDL	SDL
SDA	SDA

Biped Walking Gait



Biped Walking Strategy

Step No	LB	LM	LU	RU	RM	RB
0	90	90	90	90	90	90
1	45	90	90	90	90	70
2	45	105	105	105	105	70
2A	45	105	105	105	105	70
3	90	105	105	105	105	90
4	110	105	105	105	105	135
5	110	75	75	75	75	135
6	90	75	75	75	75	90



Result

- Finally, we made 6 – DOF Biped Walking Robot using Arduino Mega

